

Principles of Data Quality Management ISO 8000 Part 61

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INFQ 7303, Principles of Information Quality

What Does “Quality” Mean?



**Quality
Service
You Can
Count On!**



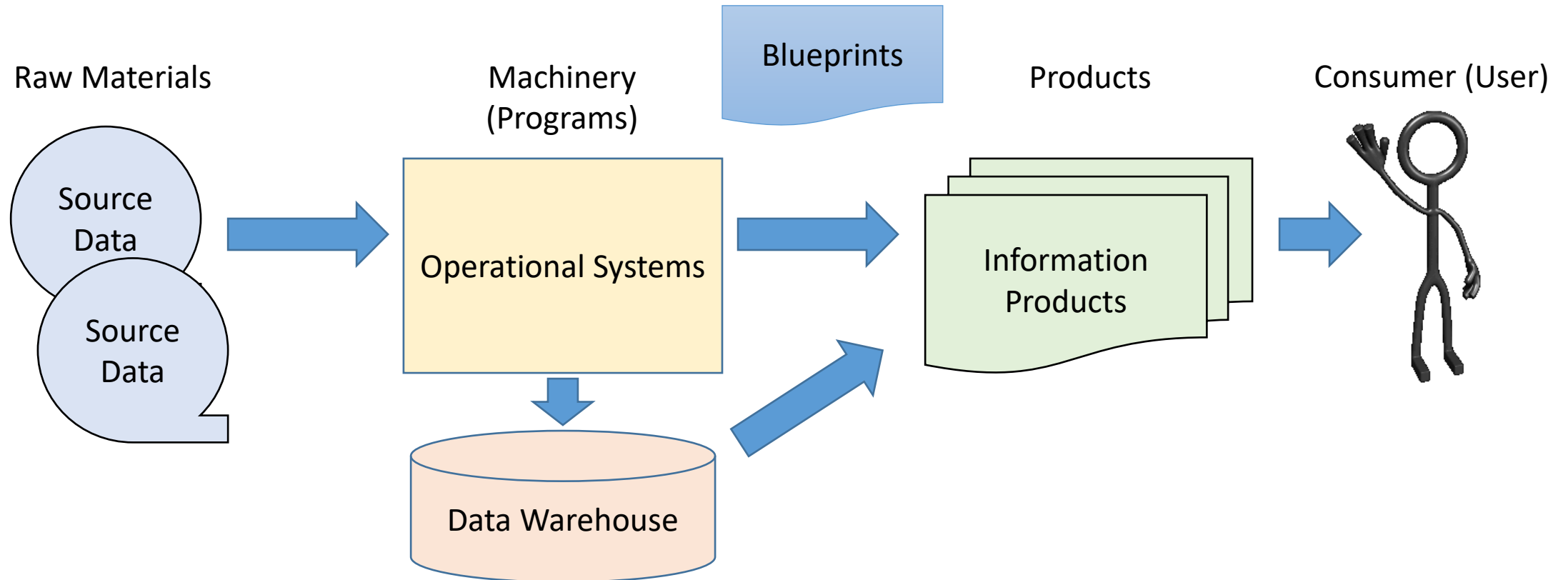
Quality is Meeting Requirements

- Quality is the
“degree to which a set of inherent characteristics fulfills requirements”
*ISO 9000:2005(E) 3.1.1
- Quality only makes sense in the context of requirements
- Data Quality is the degree to which data meet “data requirements”

Managing Information as Product

One of the Seven Basic Principles of Information Quality

Think of your organization's IT systems as an "Information Product" factory



Four things we can learn from manufacturing, How Toyota overtook GM and Ford*

To build a quality product you must first

1. Understand the consumer's needs
2. Have a well-defined production process
3. Manage the life cycle of the product
4. Appoint a product manager to translate consumer needs into product requirements

*Basic principles of Total Quality Management (TQM)

Do You Know Your Information Products?

- What are the information products being produced by your information system?
 - Reports
 - Files
 - Charts
- Information Products are part of your Data Assets
 - Are they in your Business Glossary?
- Which information products are the most important?
- Who are the users of the products? And how do they use them?



Understand the “big picture” of data quality management (DQM)

DQM has three Concurrent Processes*

1. Implementation Process
2. Data-Related Support Process
3. Resource Provisioning Process



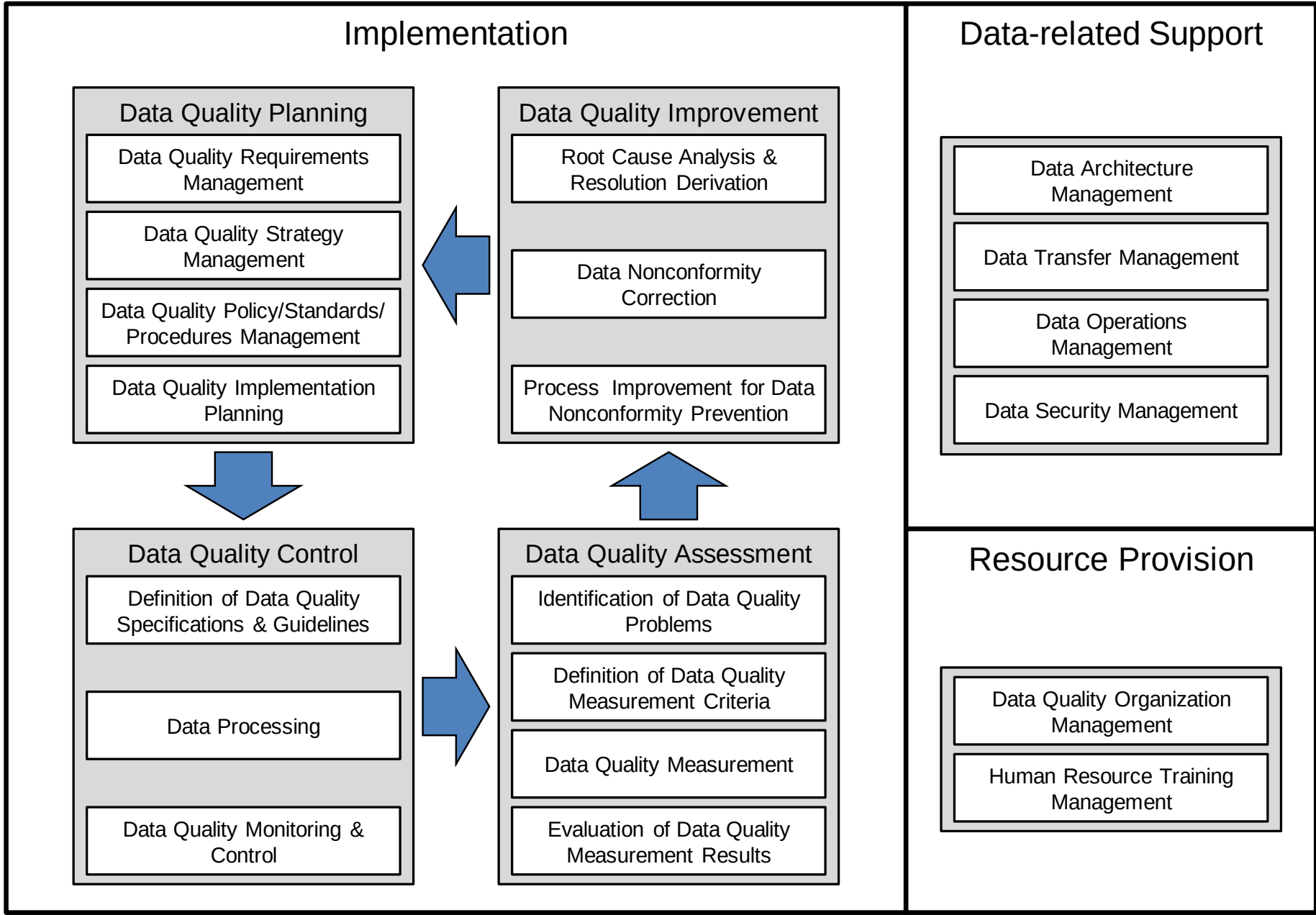
*ISO 8000-61:2016
Data Quality Management Reference Model



Implementation Process is a Four-Part Cycle

Based on the Shewhart-Deming “Plan-Do-Check-Act” Cycle

- Data Quality Planning and Requirements (Plan)
- Data Quality Control (Do)
- Data Quality Assurance (Check)
- Data Quality Analysis and Improvement (Act)



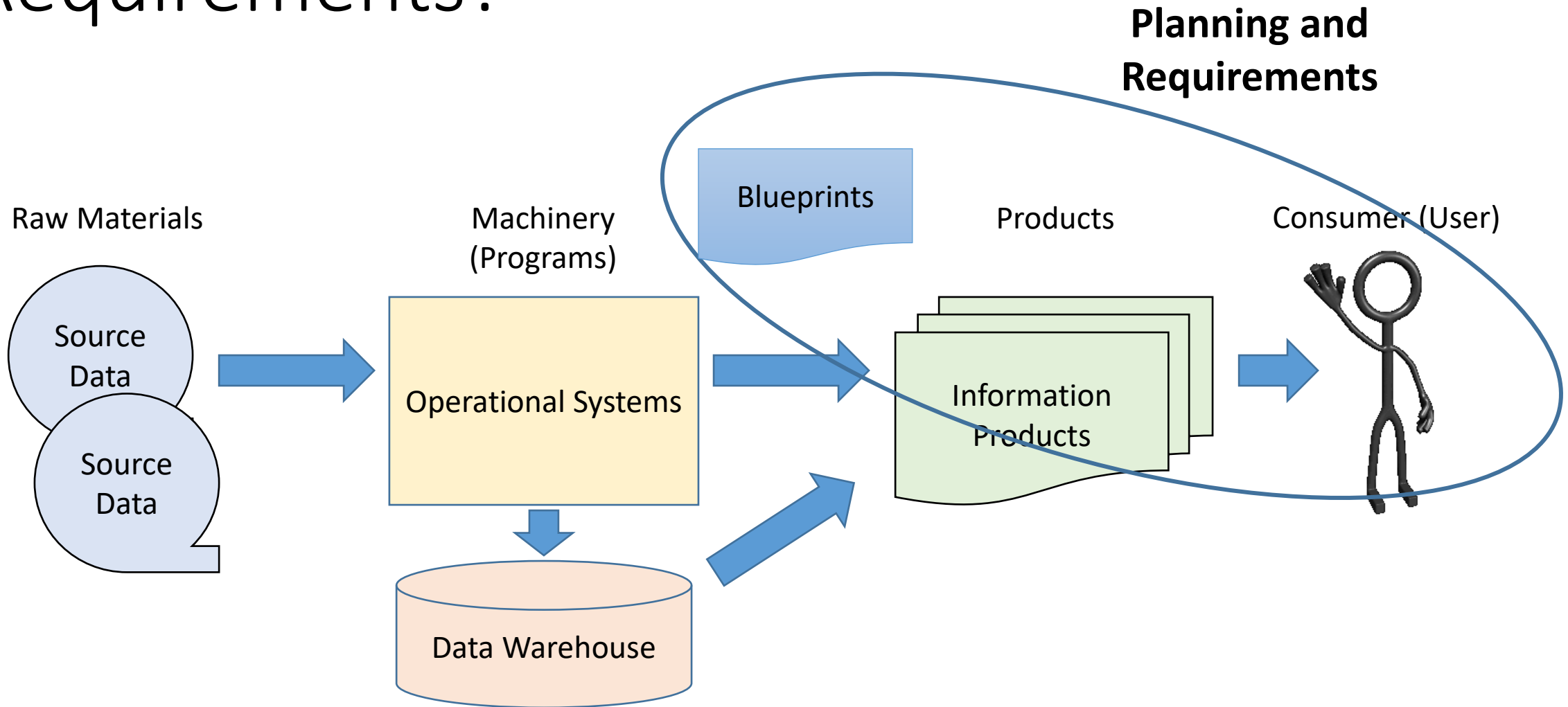
Implementation Process

Implementation Process Components

ISO 8000 Part 61: DQM Reference Model

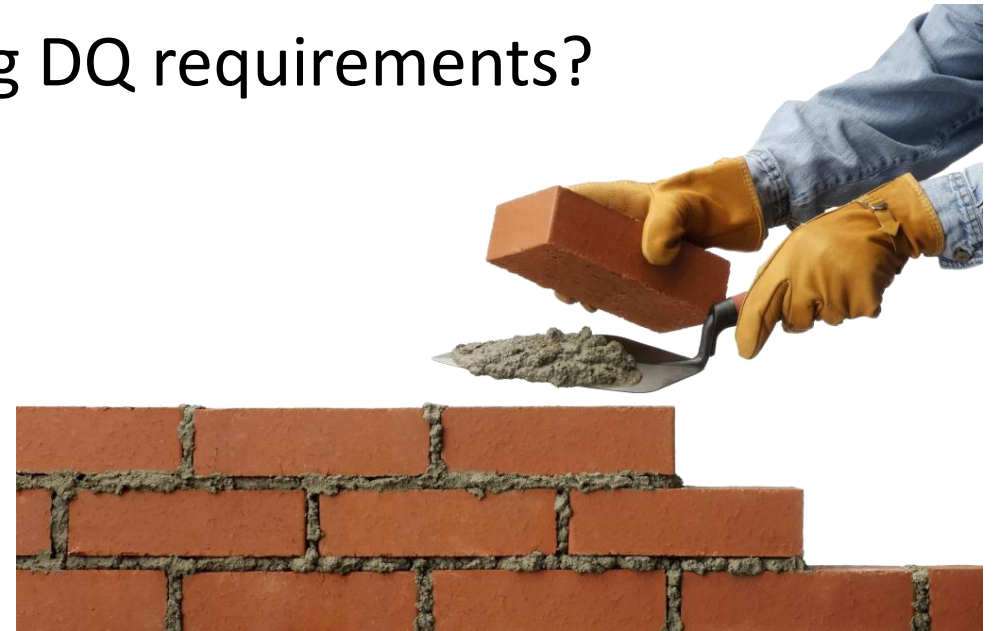
1. Data Quality Requirements and Planning
2. Data Quality Control
3. Data Quality Assurance
4. Data Quality Analysis and Improvement

Data Quality Planning: Technical and User Requirements?



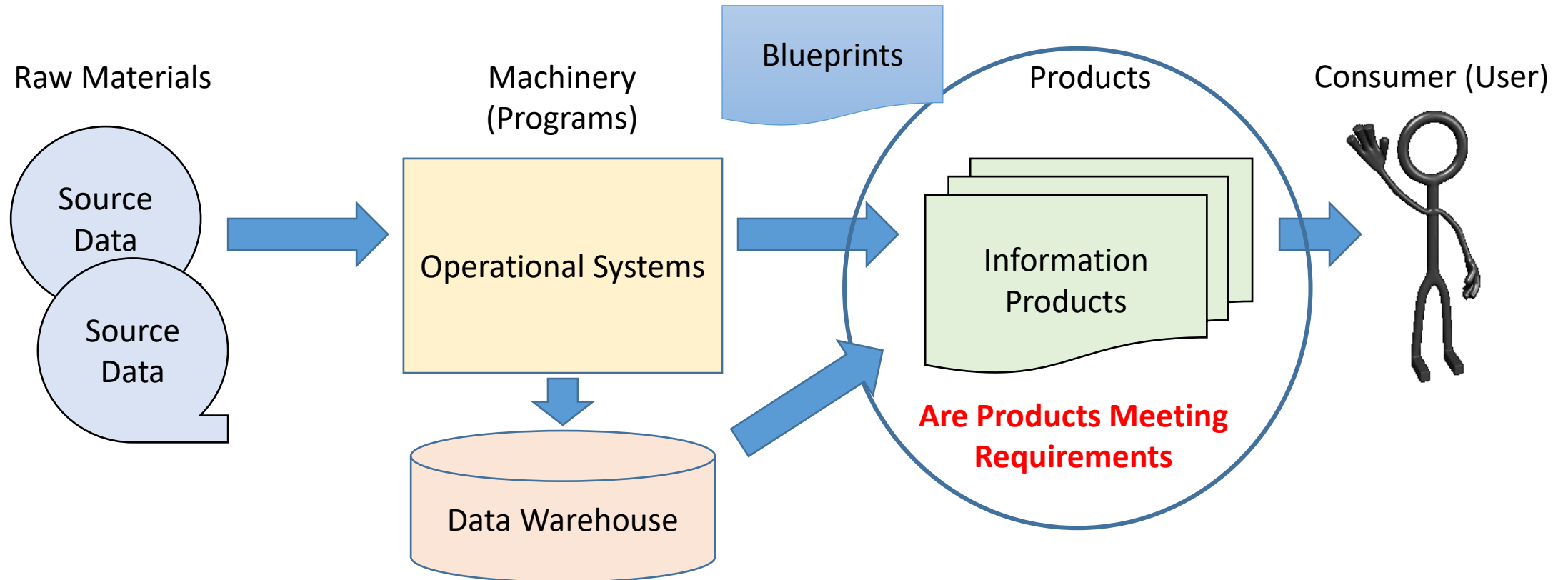
Planning and Requirements

- Do you have written documentation defining the quality requirements to be met by each information product
- Is it clear who is responsible for meeting DQ requirements?
- Do you have a strategy for improving DQ in the organization



Quality Control: Are the final products meeting (user) requirements?

Quality Control

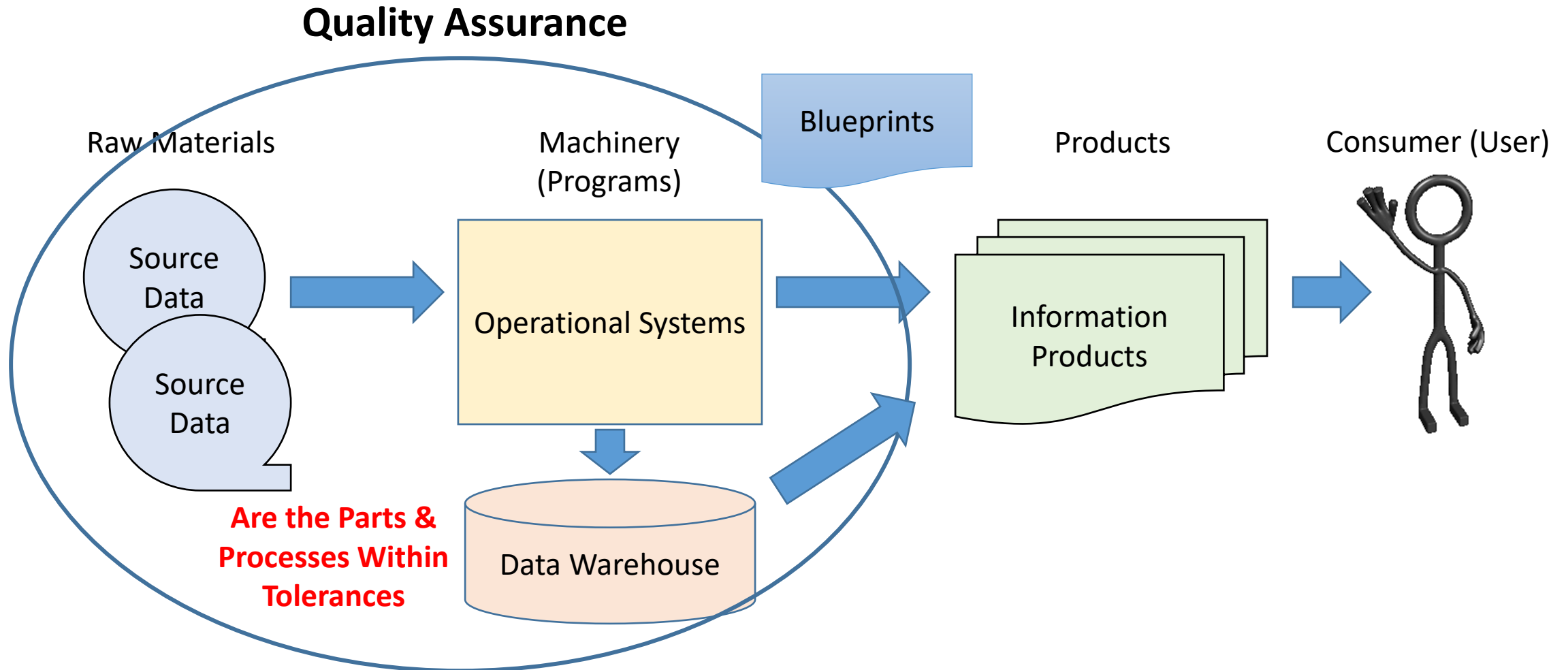


QC for Every Information Product

- Do you have a QC process in place for each product?
- Can the QC process be automated?
- If QC is manual, can you inspect 100% of products?
- If not, consider stratified sampling based on error probability indicators



Quality Assurance: Are the data & processes making the product meeting requirements?



QA Checks for All Components and Processes

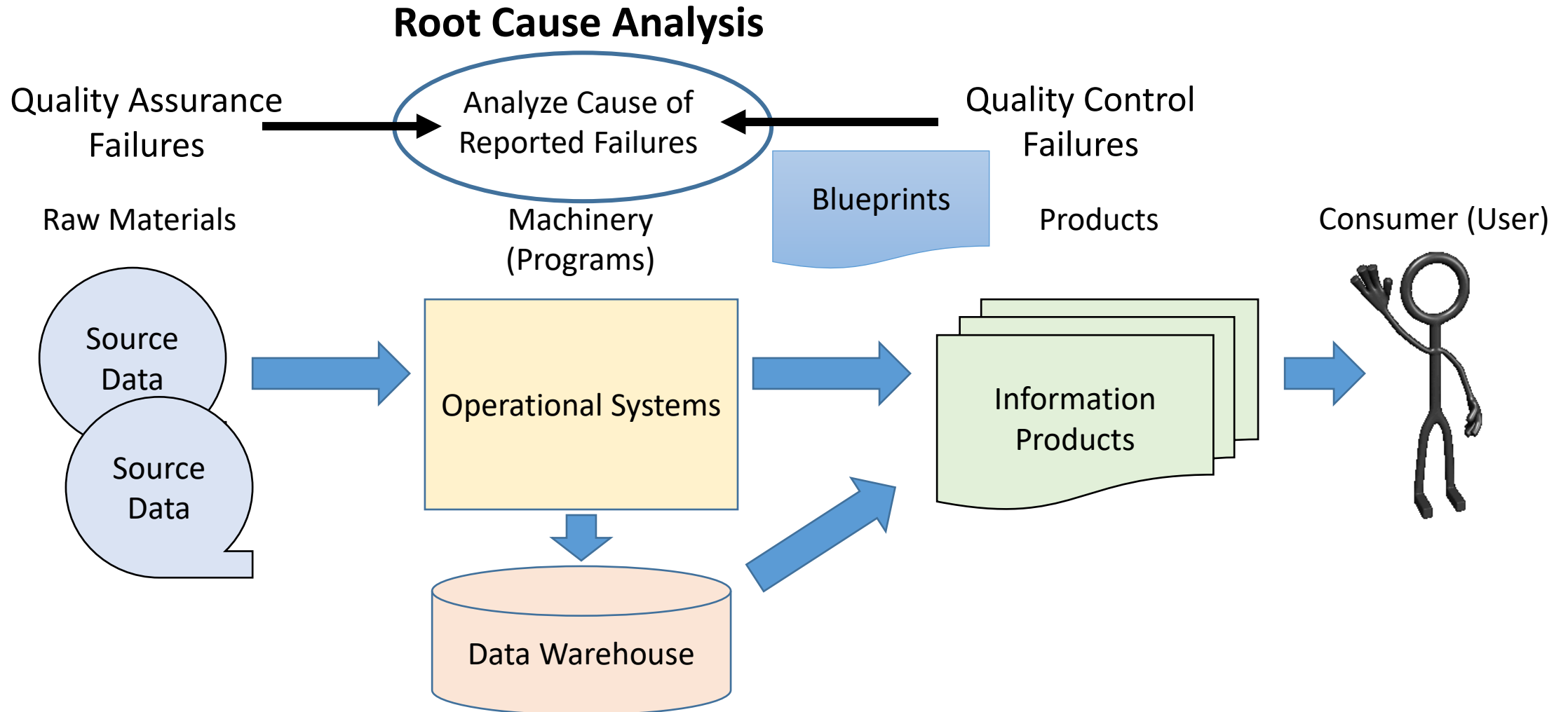
- Do you have a QA process in place for the component and processes used to make each information product?
- Can the QA processes be automated?
- If it QA is manual, can you inspect 100% of the product?
- If not, use stratified sampling based on error probability indicators



Quality Control versus Quality Assurance

- They have different meanings
- Not interchangeable
- As an IQ Profession, please use them properly
- Quality Control only applies to the finished product
Does the product meet requirements?
- Quality Assurance only applies to the process of building the product
Do the product components and product building processes meet requirements?

Root Cause Analysis: What is causing quality control and quality assurance failures?

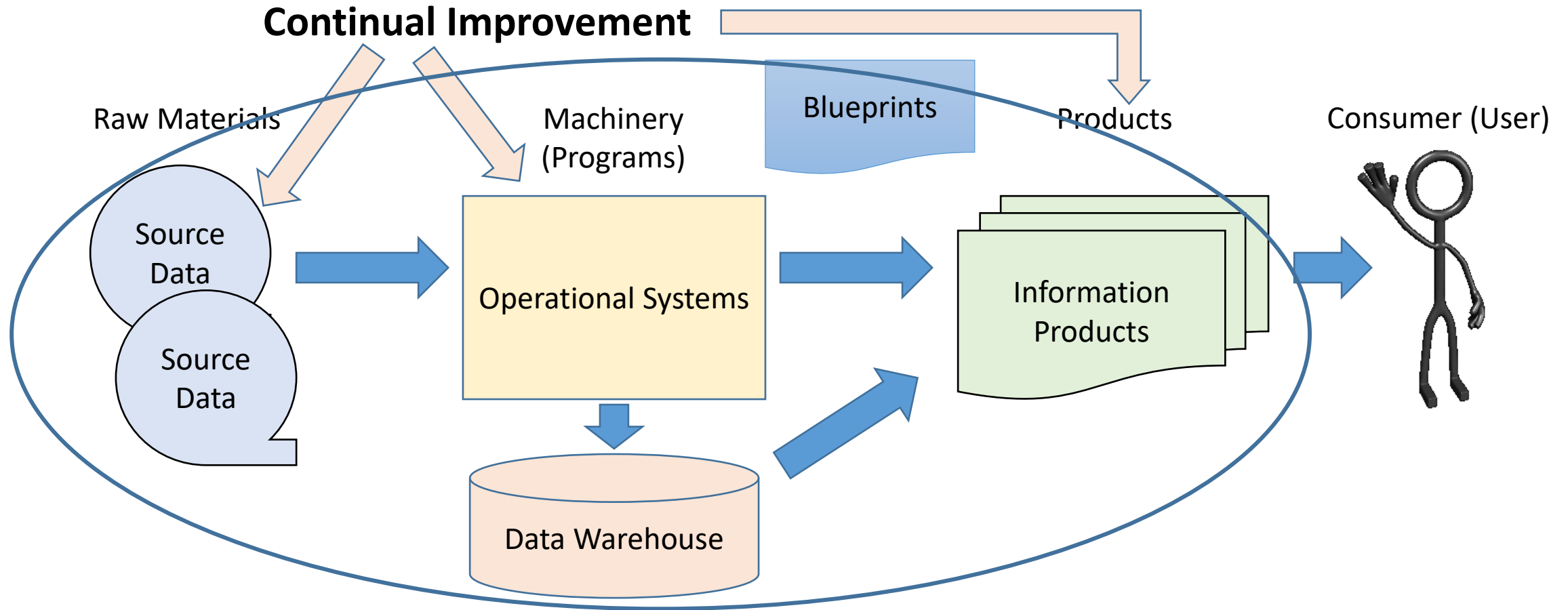


Analysis Starts with Reporting

- Implement a centralized data issue reporting system (ticket system)
- Systematically collect information on QC and QA failures
- In addition to addressing specific issues, analyze aggregate trends
- Look for root causes of persistent issues



Continual Improvement: Improve data, processes, and products based on analysis, repeat the cycle



Prioritize and implement improvements

- Prioritize improvement projects by cost vs. benefit
- Document and communicate improvements and their value
- Brag about what you do!



Data-Related Support Process

Data-Related Support Process

- Data Architecture Management
- Data Transfer Management
- Data Operations Management
- Data Security Management

Resource Provisioning Process

Resource Provisioning Process

- Data Quality Organization Management
- Human Resource Training Management

Compliance with ISO 8000-61

Prepare documentary evidence of DQM Part 61 implementation by identifying the

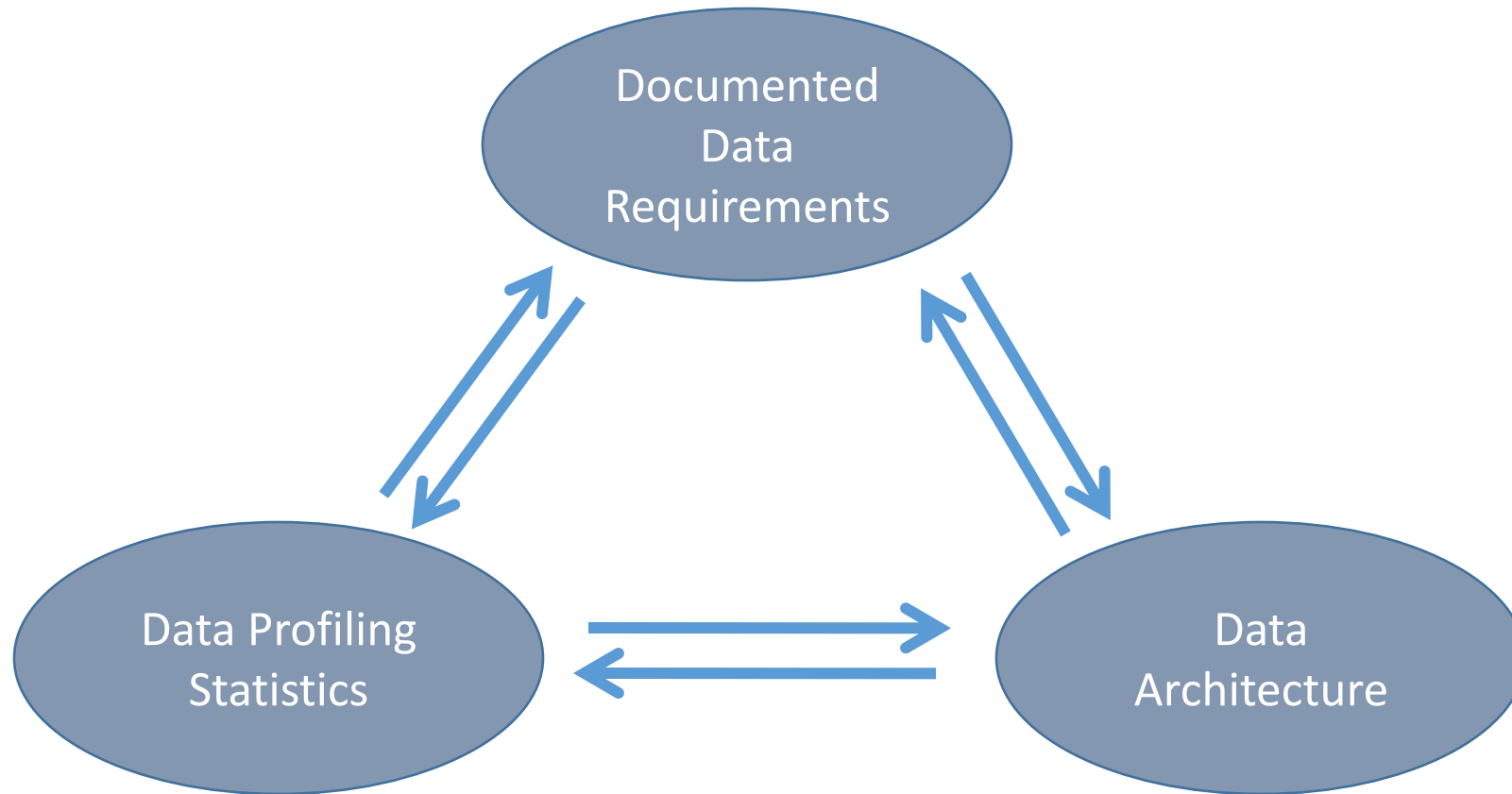
1. The activities performed and the outcomes achieved, as per clauses 6 to 8;
 - 51 Activities
 - 82 Outcomes
2. The extent to which data quality has been improved;
3. The assigned roles and responsibilities for data quality management across the organization;
4. The resources used in performing data quality management;
5. How data quality management is applied anywhere the organization is creating or using data.

Assessing Data Quality

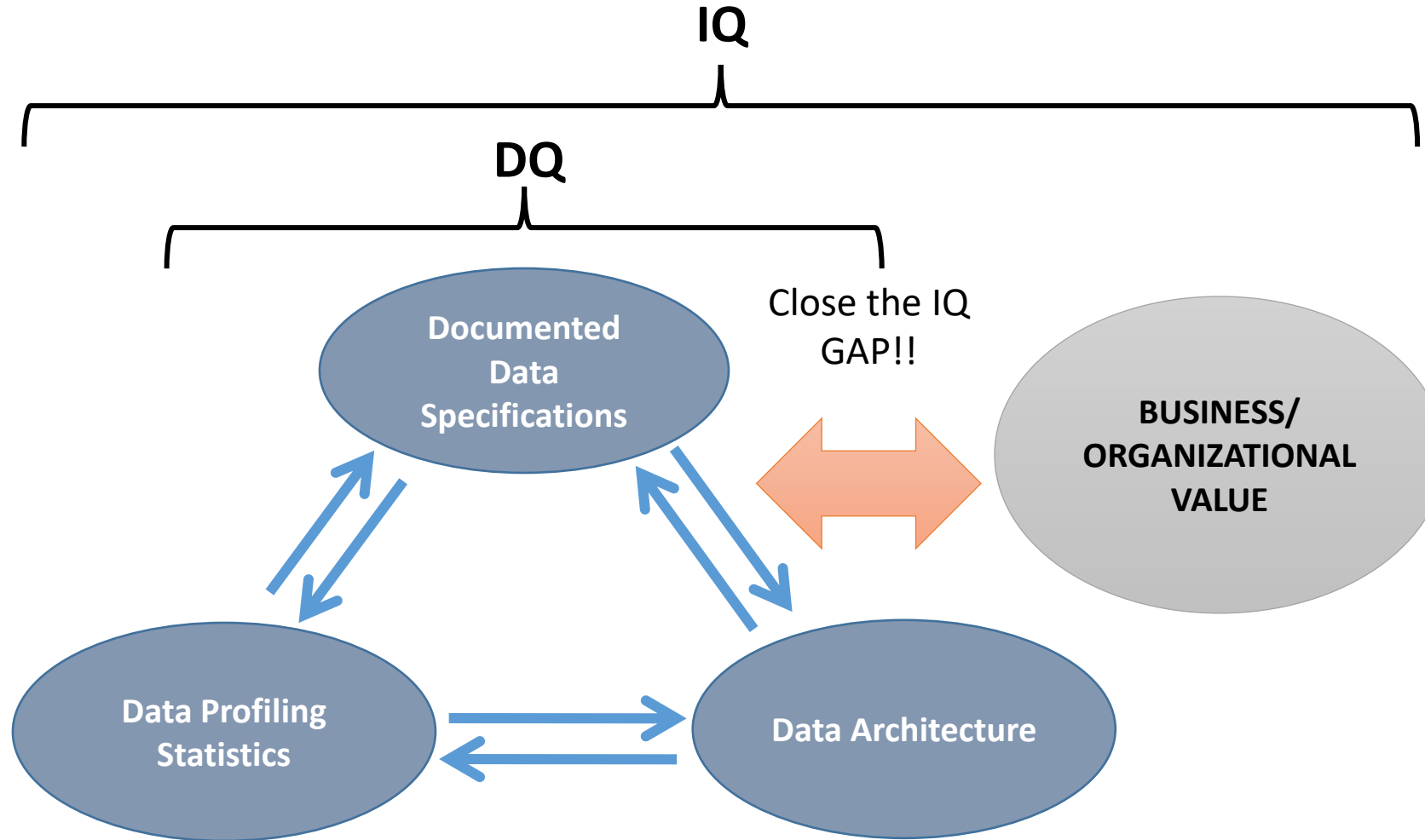
Gather and Document Requirements

- Without requirements, you can't measure and improve data quality!
- Some requirements may already be known or given
 - For example, from data governance glossary
- Some requirements may have to developed

The DQ Assessment Triangle



IQ Assessment



How are data requirements expressed?

- Data requirements are expressed in terms of data quality **dimensions** also sometimes called **characteristics**
- There are several dimensional frameworks to choose from
 - Wang-Strong Framework of Data Quality Dimensions
 - McGilvray Dimensional Framework
 - ISO/IEC 25012:2008 Characteristics
 - ...

Example Dimensional Frameworks

Wang-Strong (16)

- Accuracy
- Believability
- Objectivity
- Reputation
- Value-Added
- Relevancy
- Timeliness
- Completeness
- Amount of Data
- Interpretability
- Ease of Understanding
- Representational Consistency
- Conciseness of Representation
- Manipulability
- Access
- Security

McGilvray (12)

- Data Specifications
- Data Integrity Fundamentals
- Duplication
- Accuracy
- Consistency and Synchronization
- Timeliness and Availability
- Ease of Use and Maintainability
- Data Coverage
- Presentation Quality
- Perception, Relevance, and Trust
- Data Decay
- Transactability

ISO 25012:2008 (15)

- Accuracy
- Completeness
- Consistency
- Credibility
- Currentness
- Accessibility
- Compliance
- Confidentiality
- Efficiency
- Precision
- Traceability
- Understandability
- Availability
- Portability
- Recoverability

Start with the Basic Dimensions

- Completeness – are the data there?
- Consistent representation – are all values for the same attribute expressed in the same format and have the same meaning?
- Validity – are the values in the domain of expected values?
- Accuracy – are the values correct (different from valid)?
- Redundancy – are there unnecessary or inconsistent duplicated values?
- Timeliness – are the values available in the time needed?

Breadth of Completeness: Population Coverage

For Example

- Product Master – does it have an entry for every product
- Reference Data – complete set of building locations

Depth of Completeness: Amount of Data in Each Record

For Example

- Customer Master – does each master record have a both a current address and previous address, phone number, etc.
- How much information in each record

Completeness of Values: Problem of Missing Values

- Null values
 - Artifact of database systems, don't exist in other types of data stores
- Empty values
 - Defined fields that do not contain any data
- Blank values
 - Field contains only blank characters (or white space)
- Place holder/default values
 - N/A, None, ?, TBD, NMN, 01/01/1900

Consistent Representation: Data Standardization

- Dates
 - 05/07/87 (May 7, 1987 in U.S. but July 5, 1987 in Europe)
 - 1997-05-12 (ISO Standard, highly recommended)
- Phone Numbers
 - 501-371-7616, (501)371-7616, 501.371.7616, 5013717616
 - What about country prefix? +1 501.371.7616
 - What about extensions? 501-371-7616 x203
- Names
 - Parsed vs unparsed – “John R Talburt” vs. “John”, “R”, “Talburt”
 - Punctuation – Bailey O’Brian, Bailey OBrian, Bailey O Brian
 - Prefix and suffix titles – Dr. John R Talburt, IQCP
 - Generational suffixes – Sammy Davis, Jr.

Valid Values: (Data Validation)

Are the data values rational?

- Ranges
 - Date-of-Birth between 1900-01-01 and 1999-06-12 (At least 18 as of today)
- Set Membership
 - AccountType in {"A", "B", "C"}
 - State Code is found in the State Code Reference Master Table
- Field-to-Field
 - Termination_Date > Hire_Date

Accuracy:

Are the values actually correct?

- The most difficult dimension to assess
- Requires verification against “real world” or “system of record”
- Do not confuse “Accuracy” with “Validity”
 - Customer “John Doe” has date-of-birth (DOB) value given as “1977-12-03”
 - DOB meets required format CCYY-MM-DD
 - DOB falls in acceptable range between 1900-01-01 and 1999-06-12
 - Therefore is the a “Valid Value”
 - BUT is it accurate?
 - To know requires verification with Driver’s License, Public Health Records, ...

Accuracy and Validation are Asymmetric

- Values that fail validation are likely inaccurate
- But values that pass validation are not necessarily accurate
- In other words, some inaccurate values may pass validation

Data Redundancy:

Multiple copies of the same information

- Root cause of many data quality issues
- Easy to lose synchronization
 - For example, student name and address stored in two different places, but student's change of address is only updated in one copy- creates conflict
- Leads to over counting and under counting
 - Same customer recorded in customer master table 2 or 3 times with different name or address variations – Counting rows of table “over counts” customers
 - Same product recorded in product master under two different product codes can “under count” the total number of sales for the product

Data Profiling

- Process for examining a dataset (database or flat file) that collects statistics about data values and relationships between records
- Profiling statistics often become the “**real**” data specifications (metadata)

Look at the DATA!!!



Data Assessment is Fun!

- Bring out the detective in you
- Discovers things that other people don't know
- Let's you be creative
- More fun when you have good tools

Three-Dimensional Analysis*

- Column Profiling
- Dependency Profiling
 - Intra-table (column-to-column) dependencies
- Redundancy Profiling
 - Inter-table (table-to-table) dependencies and overlap

*Ed Lindsey – *Three-Dimensional Analysis Data Profiling Techniques*

Column Analysis

- Degree of uniqueness of value
- Frequency of unique values
- Frequency of unique patterns
- Percentage of null values
- Inferred data type
- Outlier values
 - Largest (in collation, in length)
 - Smallest (in collation, in length)
- Average value

Example Column Analysis “Field12”

Overall Statistics

Field: Field12	
Defined type: VARCHAR	
Defined length: 255 chars	
Column Profiling	Frequency Distribution
Pattern Frequency	
Metric Name	Metric Value
Ordinal Position	12
Count	10081
Null Count	324
Blank Count	0
Minimum Value	#input error
Maximum Value	unknown
Mode	USD
Pattern Count	9
Unique Count	13
Uniqueness	0.13
Primary Key Candidate	no
Data Type	VARCHAR
Data Length	255 chars
Actual Type	string
Minimum Length	1
Maximum Length	21
Mean	(not applicable)
Median	(not applicable)
Non-null Count	9757
Nullable	YES
Decimal Places	0
Standard Deviation	(not applicable)
Standard Error	(not applicable)
Percent Null	3.2

Example Column Analysis “Field12”

Field: Field12			
Defined type: VARCHAR			
Defined length: 255 chars			
Column Profiling			
Frequency Distribution			
Pattern Frequency Distribution			
Percentiles			
Outliers			
No			
Value	Count	Percentage	
USD	9723	96.45	
(null value)	324	3.21	
UDS	6	0.06	
US	6	0.06	
#input error	5	0.05	
GBP	4	0.04	
** To be confirmed **	3	0.03	
** TBC **	3	0.03	
unknown	2	0.02	
Home	1	0.01	
HKD	1	0.01	
UKP	1	0.01	
??	1	0.01	
H	1	0.01	

Value Frequency Analysis

Example Column (Pattern) Analysis “Field12”

Field: Field12		
Defined type: VARCHAR		
Defined length: 255 chars		
Column Profiling	Frequency Distribution	Pattern Frequency Distribution
Percentiles	Outlier	
Pattern	Alternate	Count
AAA	A(3)	9735
AA	A(2)	6
#aaaaa aaaaa	#a(5) a(5)	5
** AAA **	*(2) A(3) *(2)	3
** Aa aa aaaaaaaaaa **	*(2) Aa a(2) a(9) *(2)	3
aaaaaaa	a(7)	2
??	?(2)	1
Aaaa	Aa(3)	1
A	A	1

Pattern Frequency Analysis

Intra-Table Determinate Analysis

Customer ID	Name	Phone
12345	Ed	555.831.1234
34678	Tim	555.448.1234
24804	Lynn	555.567.1234
14268	Deana	555.226.1234
29966	Tim	555.678.1234
28736	Dave	555.334.1234
12345	Ed	555.831.4321

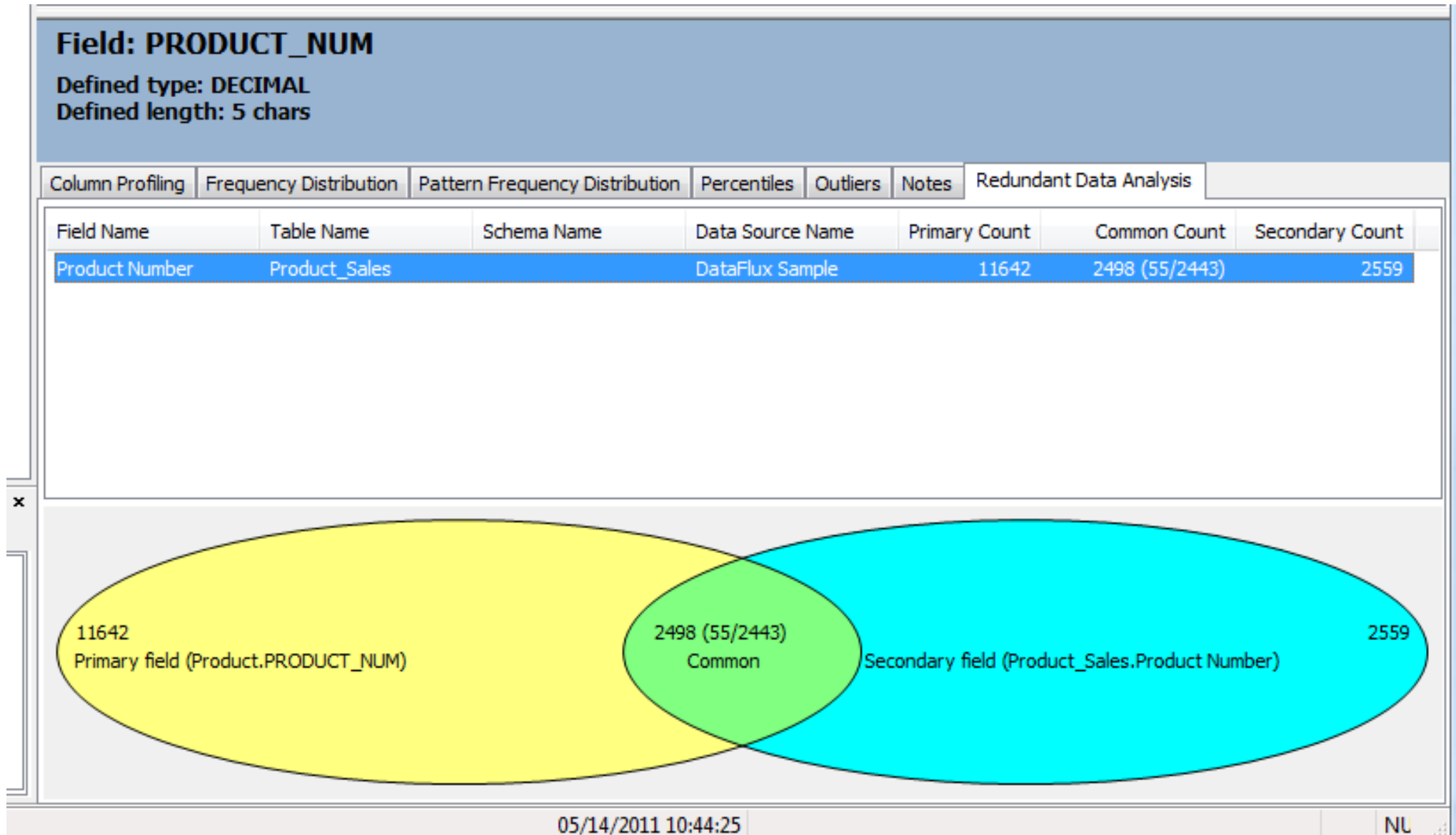
CustomerID is
determinate for
Name

CustomerID is
not determinate
for Phone

Inter-Table Analysis

- Check key and foreign key integrity
- Redundant data analysis
- Tools have varying capability in this area

Example from DataFlux Profiling

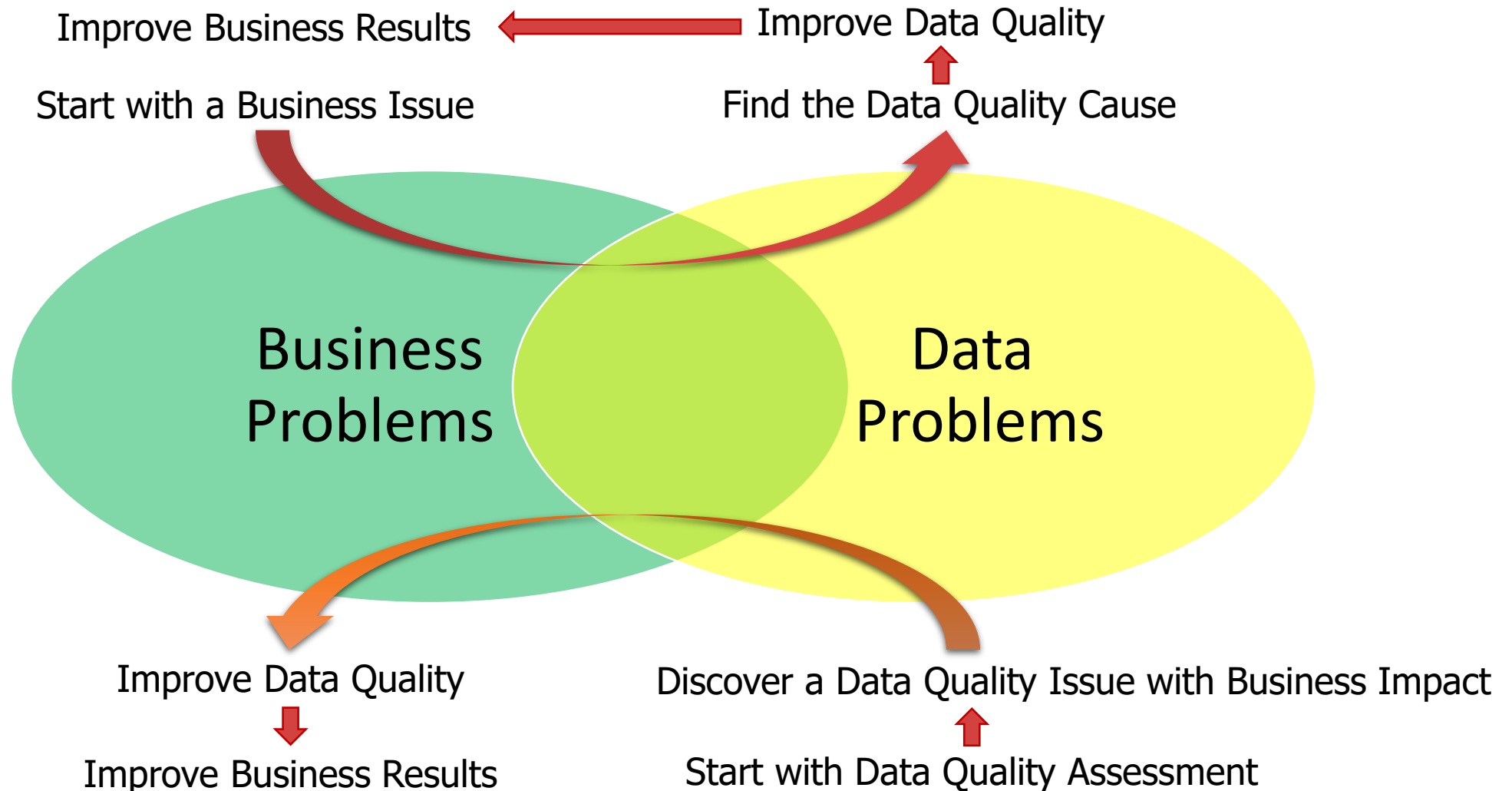


Questions ??

Getting to Value

The ROI for Data Quality

Look for data quality value both Top-Down and Bottom-Up



Places to Find Information Value

- Financial
 - Operating Costs
 - Revenue
 - Cash Flow
 - Charges
 - Budget
- Confidence
 - Trust
 - Decisions
 - Control
 - Forecasting
 - Reporting
- Risk Mgmt
 - Regulatory
 - Credit
 - Investment
 - Competitiveness
 - Leakage
- Health
 - Disease control
 - Treatments
 - Mental
 - Diet/Fitness
 - Research
- Compliance
 - Government
 - Industry
 - Privacy
- Satisfaction
 - Customer
 - Supplier
 - Employee
- Productivity
 - Workloads
 - Throughput
 - Quality
- Safety
 - Accident prevention
 -

The Business Case for Data Quality

- You must develop a convincing business case for data quality to be successful
- A sound business case will
 - Help management see data quality as an “investment” not an added expense
 - Gain support from management
 - Justify the resources necessary to be successful
- Show how the benefits of data quality are greater than the investment
- Let’s you “**sell**” data quality to management –
It should be something that they want to buy



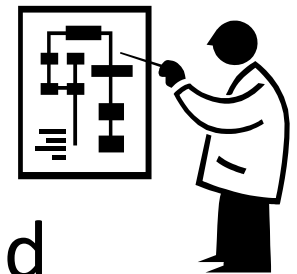
Align Data Quality with Business Goals and Listen!

- Make a clear connection between information and its quality to business goals and objectives
- Collect stories where people in the organization already suspect that
 - Poor data quality has created a business problem
 - Having higher quality data would open up new business opportunities

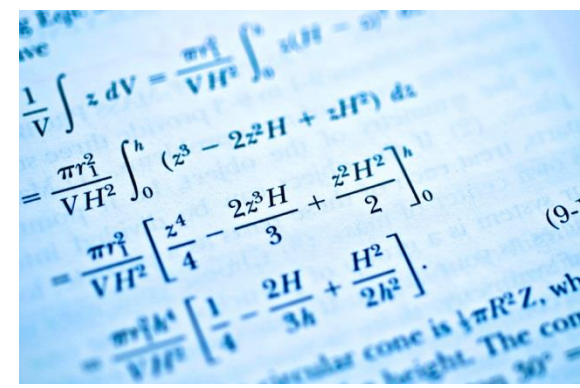


Understand the Full Context

- Understand how people, departments, processes, & technology are using data and information to create value
- Based on the data quality assessment, look at where and how data of low quality is being used
 - Remember: Value is created only when information is used
- Determine what data elements are the most critical for the organization and in which information products these elements are used
 - Critical Data Element (CDE) analysis



Be Quantitative



An analysis indicated the customer master list has 22% duplicates. A better matching algorithm would reduce this to 3% or less

- Qualitative: Improving the process to remove duplicates from the customer master do a better job of bring together customer accounts, give customers better service, and improve customer satisfaction
- Quantitative: The improve is expected to increase customer retention by 10% a year.
 - Cost to acquire new customer is \$1,200,
 - Cost to retain a customer is only \$400.
 - For 2,000 customers, savings = $0.1 * 2,000 * (1,200 - 400) = \$160,000/\text{yr}$